Maths for Machine Learning compact cheatsheet

STEP 1 School foundations

1.1 Numbers, Algebra and Equations

Why machine learning needs maths A machine learning model is a function with knobs . You give it numbers (the size of a house, the pixels of a photo, the words of a message) and it gives back a number (a price, the word "cat", the chance that a message is spam).

Kinds of numbers and how computers store them Numbers come in families. Each family is bigger than the one before and contains it.

Order of operations, fractions and percentages When an expression mixes operations, everyone must work it out in the same order, or two people would get two different answers.

Variables, expressions and equations A variable is a box with a name, such as x, that holds a number we do not know yet. An expression such as 3x + 5 is a recipe that uses the box. An equation such as 3x + 5 = 20 says two things are equal, and solving it means finding what is in the box.

Powers, roots and scientific notation A power is repeated multiplication: $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$. The small number 5 is the exponent and 2 is the base .

Sigma and Pi notation: the maths shorthand Machine learning formulas add up many terms at once. Writing $x_1 + x_2 + x_3 + \ldots + x_n$ is slow, so mathematicians use the Greek capital letter sigma, Σ, which means "sum".

Absolute value, inequalities and distance The absolute value |x| is the distance of x from zero, so it is never negative: |5| = 5 and |−5| = 5. The distance between two numbers a and b is |a − b|, whichever is bigger.

Machine epsilon
$\varepsilon = 2^{-52} \approx 2.2 \times 10^{-16}$ for float64

BODMAS (also called PEMDAS)
B → O → D·M → A·S

Two equations, two unknowns (Cramer rule)
$a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ give $x = \frac{c_1b_2 - c_2b_1}{a_1b_2 - a_2b_1}$ and $y = \frac{a_1c_2 - a_2c_1}{a_1b_2 - a_2b_1}$

Sigma means add them all
$\sum_{i=1}^{n} x_i = x_1 + x_2 + \ldots + x_n$

Absolute value
$|x| = x$ if $x \geq 0$, and $-x$ if $x < 0$

TRAP Never test two decimal results with ==. Test whether they are close: `abs(a - b) < 1e-9`. NumPy calls this `np.isclose`, and PyTorch calls it `torch.allclose`.

EXAM |x − 10| ≤ 2 means 8 ≤ x ≤ 12, not 8 ≤ x ≤ 10. Students often forget the right half of the interval.

1.2 Functions, Graphs, Exponents and Logarithms

What a function is A function is a machine with one rule: every input gives exactly one output. A vending machine is a function. Press B4 and you always get the same snack. We write the machine as f and its output for input x as f(x), read "f of x".

Straight lines: slope and intercept The simplest useful function is a straight line. Two numbers describe it completely.

Quadratics and polynomials A quadratic has an x^2 term: $f(x) = ax^2 + bx + c$. Its graph is a parabola , a U shape when a > 0 and an upside down U when a < 0. The bottom (or top) is called the vertex .

Exponential growth and the number e In an exponential function the variable sits in the exponent: $f(x) = a^x$. Each step multiplies instead of adding. With a > 1 it grows faster and faster, with 0 < a < 1 it shrinks towards zero but never reaches it.

Logarithms: the undo button for powers A logarithm answers the question "what power do I need?". Since $2^3 = 8$, $\log_2 8 = 3$. Since $10^2 = 100$, $\log_{10} 100 = 2$. The natural log, ln, uses base e.

The S curve and other functions machine learning uses Some functions show up in machine learning again and again. They are called activation functions because they decide how strongly a neuron "fires".

Composing, shifting and inverting functions Composition feeds one function into another: $(f \circ g)(x) = f(g(x))$. Work from the inside out. If g(x) = 3x − 1 and f is the sigmoid, then f(g(1)) = σ(2) ≈ 0.88.

Function notation
$f(x) = 2x + 1 \implies f(3) = 2 \times 3 + 1 = 7$

Slope intercept form $y = m\,x + b$

Quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Euler number
$e = \lim (1 + \frac{1}{n})^n$ as $n \to \infty \approx 2.71828$

Definition $\log_b(y) = x$ means $b^x = y$

Sigmoid $\sigma(z) = \frac{1}{1 + e^{-z}}$

Composition $(f \circ g)(x) = f(g(x))$

EXAM ln(0) does not exist and ln of a negative number is not real. Code adds a tiny epsilon, as in log(p + 1e-12), so a predicted probability of exactly 0 does not crash training.

STEP 2 Linear algebra

2.1 Vectors

What a vector is A vector is an ordered list of numbers. You can picture it in two ways, and both are useful.

Adding, subtracting and scaling vectors Vectors are added part by part: [3, 1] + [1, 2] = [4, 3]. As arrows, put the tail of the second at the tip of the first. The sum goes from the very start to the very end. Multiplying by a number (a scalar) stretches the arrow: 2 × [3, 1] = [6, 2].

Length (norm) and distance The length of a vector is called its norm . The usual one comes straight from Pythagoras: [3, 4] has length $\sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$. There are other useful ways to measure size.

The dot product The dot product turns two vectors into one number. Multiply matching parts, then add: [2, 3] · [4, −1] = 2 × 4 + 3 × (−1) = 8 − 3 = 5. It also has a meaning in pictures: it measures how much two arrows point the same way.

Cosine similarity: how alike are two things If you divide the dot product by both lengths, you get cos θ alone. This is cosine similarity .

Projection: the shadow of a vector Shine a light straight down onto a line. The shadow a vector casts on that line is its projection . It tells you how much of one vector lies along the direction of another.

Span, independence and basis Take all the linear combinations of some vectors. Every point you can reach is called their span . With [1, 0] and [0, 1] you can reach any point in the plane: [5, −2] = 5·[1, 0] − 2·[0, 1].

Writing a vector
$v = [v_1, v_2, \ldots, v_n]$ or as a column $\begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$

Addition $a + b = [a_1 + b_1, a_2 + b_2, \ldots]$

L2 norm (Euclidean)
$\|v\|_2 = \sqrt{(v_1^2 + v_2^2 + \ldots + v_n^2)}$

Algebra form
$a \cdot b = a_1b_1 + a_2b_2 + \ldots + a_nb_n = \Sigma\, a_ib_i$

Cosine similarity $\cos(a, b) = \frac{a \cdot b}{\|a\|\,\|b\|}$

Scalar projection (shadow length)
$\text{comp_b}\ a = \frac{a \cdot b}{\|b\|}$

Basis and dimension
basis = a set of independent vectors that spans the space

TRAP Distances are only fair when features use similar scales. If one feature is salary in rupees (around 500000) and another is age in years (around 30), salary dominates every distance. Scale the features first, for example to a mean of 0 and a standard deviation of 1.

2.2 Matrices and Matrix Multiplication

What a matrix is A matrix is a rectangle of numbers arranged in rows and columns. Its shape is written rows × columns, so a matrix with 2 rows and 3 columns is 2 × 3. The entry in row i and column j is written a_{ij}.

Adding, scaling and transposing Matrices of the same shape add entry by entry. Multiplying by a number scales every entry. The transpose A^T flips a matrix over its diagonal, so rows become columns and a 2 × 3 matrix becomes 3 × 2.

Matrix multiplication, step by step To multiply A by B, each entry of the answer is a dot product : row i of A with column j of B. Walk along the row and down the column together, multiply the pairs, and add.

A matrix is a transformation Multiplying a vector by a matrix moves it. Every 2 × 2 matrix stretches, squashes, rotates, shears or flips the whole plane, and keeps straight lines straight. That is why it is called a linear transformation .

Identity, determinant and inverse The identity matrix I has 1s on its diagonal and 0s elsewhere. It does nothing: AI = IA = A, just like multiplying by 1. The inverse A^{-1} undoes A: $A^{-1}A = I$.

Solving many equations at once A system of linear equations is one matrix equation Ax = b . The matrix A holds the numbers in front of the unknowns, x is the vector of unknowns, and b is the right hand side.

Rank and special matrices The rank of a matrix is the number of truly independent columns (or rows, it is always the same number). It says how many dimensions the matrix really uses. A 1000 × 50 data matrix with rank 30 has 20 columns that are combinations of others.

Addition $(A + B)_{ij} = a_{ij} + b_{ij}$

Matrix product $(AB)_{ij} = \Sigma_k\, a_{ik}\, b_{kj}$

2 × 2 determinant

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

Linear regression in matrix form (normal equations)

$$\beta = (X^T X)^{-1} X^T y$$

Rank and the lost directions

rank + (number of lost directions) = number of columns

EXAM In NumPy, `A * B` is the element wise product and `A @ B` is matrix multiplication. Mixing them up gives wrong answers without any error message when the shapes happen to fit.

TRAP If det A = 0 the matrix squashes the plane flat onto a line or a point. Information is lost, so nothing can undo it. Such a matrix is called SINGULAR and has no inverse.

2.3 Eigenvalues, SVD and PCA

Eigenvectors: the directions a matrix keeps Chapter 4 showed that a matrix moves every arrow in the plane. Most arrows get turned to a new direction. A few special arrows only get longer, shorter or flipped , and stay on their own line.

Finding eigenvalues by hand

Rules that save time Eigenvalues come with a handful of rules that let you check your work in seconds and that explain why they are so useful.

Power iteration and PageRank Real matrices have thousands or millions of rows, so nobody solves a characteristic equation. A simple and famous method is power iteration : start with any vector, multiply by A, scale it back to length 1, and repeat.

Singular value decomposition (SVD) Eigenvectors need a square matrix, and even then they might not exist. The SVD works for every matrix of any shape. It says each matrix is three simple moves in a row: a rotation, a stretch along the axes, and another rotation.

Principal component analysis (PCA) PCA finds the directions in which your data spreads out the most. Keep the top few and you can describe 100 features with, say, 10 numbers, while keeping most of the information. It is eigenvectors applied to data.

Where machine learning uses all this Eigenvalues and the SVD appear wherever a model needs the "main directions" of something. Here is a quick map, which later chapters revisit.

The eigen equation $A\,v = \lambda\,v \quad \text{with } v \neq 0$

Characteristic equation $\det(A - \lambda I) = 0$

Diagonalisation (eigendecomposition)

$$A = P\,D\,P^{-1} \quad \text{and} \quad A^n = P\,D^n\,P^{-1}$$

Power iteration

$$v \leftarrow A\,v \,/\, \|A\,v\| \qquad \lambda \approx v^T A\,v$$

Singular value decomposition $A = U\,\Sigma\,V^T$

Covariance of two features

$$\operatorname{cov}(x, y) = \frac{\Sigma\,(x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Condition number $\kappa(A) = \sigma\max / \sigma\min$

TRAP Scale your features first. If height is in millimetres and weight in tonnes, the millimetre column has a huge variance only because of its unit, and PCA will think it is the most important direction. Standardise each column (subtract the mean, divide by the standard deviation) before PCA.

STEP 3 Calculus

3.1 Derivatives and Integrals

Rate of change and slope A derivative answers one question: how fast is something changing right now? A car that travels 120 km in 2 hours has an average speed of 60 km/h, but the speedometer shows the speed at this exact moment. The derivative is the speedometer.

Limits and the derivative from first principles A limit says what value a formula approaches as the input gets close to some number, even if you cannot plug that number in. You cannot set h = 0 in (f(a + h) − f(a)) ÷ h because that divides by zero, but you can ask what happens as h shrinks.

The rules of differentiation Nobody works from first principles every time. A small set of rules covers almost every function you will meet in machine learning.

The chain rule Most functions in machine learning are functions inside functions. A neuron computes z = wx + b, then a = σ(z), then a loss L(a). The chain rule says how a small change at the start ripples through every link.

Peaks, valleys and the second derivative Where a smooth curve reaches a peak or the bottom of a valley, it is flat for a moment, so its slope is 0. Setting f'(x) = 0 finds these critical points . The second derivative f''(x), the slope of the slope, tells which kind it is.

Integrals: adding up tiny pieces An integral is the opposite job of a derivative. A derivative turns distance into speed. An integral turns speed back into distance, by adding up "speed × tiny time" over and over. In pictures, it is the area under a curve .

Why derivatives make machines learn Training a model means making a loss number small. The derivative of the loss with respect to a weight says which way to nudge that weight and how strongly. If ∂L/∂w is positive, increasing w raises the loss, so decrease it. If it is negative, increase w.

Average rate of change (slope of a secant)

$$\frac{f(b) - f(a)}{b - a}$$

Power rule $d/dx\; x^n = n\,x^{n-1}$

Chain rule

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Convex function $f''(x) \geq 0$ everywhere

Definite integral as a sum of strips

$$\int_a^b f(x)\,dx \approx \Sigma\, f(x_i)\,\Delta x$$

TRAP Do not make h too tiny on a computer. With $h = 10^{-15}$ the subtraction f(x + h) − f(x) loses almost all its digits to rounding. Something like $h = 10^{-5}$ is a safe middle ground.

3.2 Gradients and Backpropagation

Partial derivatives: one knob at a time Real things depend on many inputs. A house price depends on its size and its age. A model loss depends on millions of weights. A partial derivative asks how the output changes when you turn one input and hold every other input still.

The gradient: the steepest way uphill Collect all the partial derivatives into one vector and you get the gradient , written ∇f (said "grad f" or "del f"). It is the most important object in machine learning.

Directional derivatives and linear approximation The gradient answers "how fast does f change if I step in any direction u?" Just take a dot product. And near a point, every smooth surface looks almost flat, so a tangent plane predicts nearby values.

Jacobian and Hessian matrices Two matrices organise all the derivatives of many inputs at once.

The chain rule on a computational graph Any formula can be drawn as a computational graph : input boxes flow through small operations (add, multiply, sigmoid) to the output. The forward pass computes values left to right.

Backpropagation through a neuron Now put it all together for one neuron. Inputs x = [1, 2], weights w = [0.5, −0.3], bias b = 0.1, a sigmoid, and the squared error loss against the target y = 1.

Gradients for whole datasets A real loss is the average over many examples, so its gradient is the average of the per example gradients. With matrices, the whole batch is done in one line.

Partial derivative

$$\frac{\partial f}{\partial x} = \text{slope of f when only x moves}$$

Gradient

$$\nabla f = \begin{pmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \\ \vdots \\ \partial f/\partial x_n \end{pmatrix}$$

Directional derivative $D_u\, f = \nabla f \cdot u$

Jacobian (first derivatives of many outputs)

$$J = \begin{pmatrix} \partial f_1/\partial x_1 & \partial f_1/\partial x_2 \\ \partial f_2/\partial x_1 & \partial f_2/\partial x_2 \end{pmatrix}$$

Multivariable chain rule

$$\frac{\partial L}{\partial x} = \Sigma_{\text{paths}} \text{ (product of local slopes along the path)}$$

Mean squared error for linear regression

$$L(w) = \frac{1}{n} \|Xw - y\|^2$$

3.3 Gradient Descent and Optimisation

Gradient descent: walk downhill Training a model means finding the weights that make the loss as small as possible. Gradient descent does it the way a skier finds the bottom of a valley in fog: feel the slope under your feet, take a small step downhill, and repeat.

The learning rate: not too small, not too big The learning rate is the most important setting in training. Too small and training crawls. Too big and every step overshoots the valley floor, bouncing from side to side or flying off completely.

Stochastic and mini batch gradient descent The true gradient averages over every training example. With a million examples, that is a million calculations for one step. Stochastic gradient descent (SGD) estimates the gradient from a small random sample instead, and takes many more, cheaper steps.

Momentum, RMSProp and Adam Plain gradient descent struggles in long narrow valleys. The steep walls make it zig zag while the gentle floor makes it crawl.

Convexity, local minima and saddles For a convex loss there is one valley, so gradient descent with a sensible learning rate always reaches the best answer. Linear regression, logistic regression and SVMs have convex losses.

Optimising with constraints: Lagrange multipliers Sometimes the answer must obey a rule: a fixed budget, a fixed length of fence, probabilities that add to 1.

Training tricks: schedules, clipping and stopping Real training runs add a few simple tools around the optimiser.

The gradient descent update

$$w \leftarrow w - \eta \nabla L(w)$$

Stability rule for a bowl L = ½ a w²

$$w \leftarrow (1 - \eta a)\, w \quad \text{converges when } 0 < \eta < \frac{2}{a}$$

Mini batch gradient

$$g = \frac{1}{B} \Sigma_{i \in \text{batch}} \nabla L_i(w)$$

Momentum $v \leftarrow \beta v + \nabla L \quad w \leftarrow w - \eta v$

Condition number of the loss

$\kappa = \lambda\text{max} \div \lambda\text{min}$ of the Hessian

Lagrange condition

$\nabla f = \lambda \nabla g$ together with $g(x) = c$

Cosine decay

$$\eta(t) = \eta_{\max} \times \tfrac{1}{2} (1 + \cos(\pi\, t \div T))$$

STEP 4 Probability and statistics

4.1 Probability and Bayes' Theorem

Chance, outcomes and events Probability is a number from 0 to 1 that says how likely something is. 0 means impossible, 1 means certain, and 0.5 means as likely as not. The list of everything that could happen is the sample space , and any group of outcomes you care about is an event .

AND, OR and independence Two rules combine simple probabilities into bigger ones. OR means "at least one happens". AND means "both happen".

Counting: permutations and combinations When outcomes are equally likely, probability is just counting. Three tools do most of the counting.

Conditional probability A conditional probability answers "how likely is A, now that I know B happened?" Knowing B shrinks the world down to just the outcomes where B is true.

Bayes' theorem: updating beliefs The theorem of Bayes turns P(B | A) around into P(A | B). It is the rule for updating what you believe when new evidence arrives: start with a prior , see evidence, and get a posterior .

Naive Bayes: a classifier from counting Naive Bayes uses the theorem of Bayes to classify, for example spam or not spam. It makes one bold (naive) simplification: given the class, every word is independent of the others.

Probability inside machine learning Most models do not just say "cat". They say "cat with probability 0.92". Probability is how machine learning expresses uncertainty.

Equally likely outcomes

$$P(A) = \frac{\text{number of outcomes in A}}{\text{total number of outcomes}}$$

Addition rule (OR)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Factorial $n! = n \times (n-1) \times \ldots \times 2 \times 1$

Conditional probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Bayes' theorem

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B)} \quad \text{with} \quad P(B) = P(B \mid A)P(A) + P(B \mid \text{not } A)P(\text{not } A)$$

Naive Bayes score

$$\text{score(class)} = P(\text{class}) \times \Pi\, P(\text{word}_i \mid \text{class})$$

Monte Carlo estimate

average of f(random samples) → true average as samples grow

TRAP Mutually exclusive events are NOT independent. If A happens, B cannot, so knowing A tells you a lot about B.

TRAP P(A | B) and P(B | A) are different numbers. Most people who are fluent in French have been to France, but most people who have been to France are not fluent in French. Mixing them up is called the PROSECUTOR FALLACY .

4.2 Random Variables and Distributions

Random variables, PMF, PDF and CDF A random variable is a number that comes out of a chance event: the face of a die, the number of heads in ten flips, or the time until the next bus. We write it with a capital letter such as X, and a value it can take with a small letter such as x.

Expected value and variance The expected value E[X] (also called the mean μ) is the long run average of X if you could repeat the experiment forever.

Bernoulli and binomial: counting successes A Bernoulli trial has two results: success (1) with chance p or failure (0) with chance 1 – p. A coin toss, a free throw and "did the user click?" are all Bernoulli trials.

Poisson, uniform and exponential The Poisson distribution counts how many events happen in a fixed stretch of time or space when they arrive at random at an average rate λ (lambda). Calls to a help desk per hour, typos per page and goals per match are classic examples.

The normal distribution and z-scores The normal (or Gaussian) distribution is the famous bell curve. It is set by two numbers: the mean μ, which places the centre, and the standard deviation σ, which sets the width. Heights, measurement errors and exam scores are often close to normal.

The central limit theorem The central limit theorem (CLT) explains why the bell curve is everywhere. Take many independent random values from almost any distribution (it only needs a finite variance), and their average is close to normally distributed.

Covariance, correlation and distributions in ML So far each variable lived alone. Covariance asks whether two variables move together. If X is above its mean when Y is above its mean, the covariance is positive. Correlation rescales covariance to a number between –1 and 1 so it is easy to read.

PMF (discrete)

$$p(x) = P(X = x) \quad \text{with} \quad \Sigma\, p(x) = 1$$

Expected value $E[X] = \mu = \Sigma\, x \cdot p(x)$

Bernoulli(p)

$$P(X = 1) = p, \quad P(X = 0) = 1 - p$$

Poisson(λ)

$$P(X = k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

Normal PDF

$$f(x) = \frac{1}{\sigma \sqrt{(2\pi)}} \cdot e^{-(x-\mu)^2 \div (2\sigma^2)}$$

Central limit theorem

$$\bar{X} = \frac{X_1 + X_2 + \ldots + X_n}{n} \quad \text{is roughly} \quad N(\mu,\ \sigma^2 \div n)$$

Sample covariance

$$\text{Cov}(X, Y) = \frac{\Sigma\, (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

TRAP For a continuous variable, P(X = exactly 1.5) = 0, because a single point has no width and so no area. Only ranges have probability. That is also why a PDF can be taller than 1: its height is a density, not a probability.

TRAP Correlation is not causation. Ice cream sales and sunburn are correlated because both rise in hot weather. Also, r only measures straight line relationships: for x = −2, −1, 0, 1, 2 and y = x², r is exactly 0 even though y depends completely on x.

4.3 Statistics and Inference

Describing data: centre, spread and outliers Statistics turns a pile of numbers into a few that tell the story. The first questions are always: where is the centre, and how spread out is the data?

Samples, bias and standard error We rarely measure everyone. The whole group we care about is the population . The part we actually measure is the sample . A number worked out from the sample (such as $\bar{x}$) is a statistic , and it is our estimate of the population parameter (such as μ).

Confidence intervals A single estimate such as "18.2 hours" hides its uncertainty. A confidence interval gives a range: estimate ± margin of error. A 95% confidence interval is built by a method that catches the true value in 95% of samples.

Hypothesis tests and p-values A hypothesis test asks whether the data is surprising enough to doubt a default claim. The default, "nothing special is going on", is the null hypothesis H_0 . The claim we are testing for is the alternative H_1 .

A/B testing An A/B test is a hypothesis test that companies run every day. Visitors are split at random between the current version A and a new version B, and we compare a rate such as the share who buy.

Maximum likelihood estimation Maximum likelihood answers "which parameter value makes the data we saw most probable?" It is the idea behind how most machine learning models are trained.

Regression, R^2 and the bias and variance trade-off Linear regression fits the straight line $\hat{y} = a + b\,x$ that makes the squared vertical misses (residuals) as small as possible. It is both a statistics tool and the simplest machine learning model.

Mean $\bar{x} = \dfrac{x_1 + x_2 + \ldots + x_n}{n}$

Standard error of a mean $SE = \dfrac{s}{\sqrt{n}}$

Confidence interval for a mean
$\bar{x} \pm z^* \cdot SE$ with $z^* = 1.645$ (90%), 1.96 (95%), 2.576 (99%)

Test statistic (z-test for a mean)
$z = \dfrac{\bar{x} - \mu_0}{\sigma \div \sqrt{n}}$

Pooled rate $\hat{p} = \dfrac{\text{buys A} + \text{buys B}}{\text{visitors A} + \text{visitors B}}$

Likelihood of coin flips $L(p) = p^h\,(1-p)^{n-h}$

Least squares line
$b = \dfrac{\Sigma\,(x_i - \bar{x})(y_i - \bar{y})}{\Sigma\,(x_i - \bar{x})^2} \qquad a = \bar{y} - b\,\bar{x}$

TRAP A bigger sample shrinks random error but does nothing about BIAS . Polling only people who answer landline phones, or training a face detector only on photos of one group of people, gives a confident answer to the wrong question.

TRAP "95% confident" does not mean there is a 95% chance that μ is inside this particular interval. μ is fixed, the interval is what varies. It means that 95% of intervals made this way catch μ. The animation shows exactly that: about 1 in 20 intervals miss.

TRAP A p-value is NOT the probability that H_0 is true. And "not significant" does not prove there is no effect. It only says this data was not enough to tell.

TRAP Decide the sample size before the test and do not stop the moment p dips below 0.05. Peeking every day and stopping at the first "win" makes false positives far more common than 5%.

STEP 5 Maths inside ML models

5.1 Entropy, Loss Functions and Metrics

Information and entropy Some news tells you a lot, some tells you almost nothing. "The sun rose today" is no surprise. "It snowed in the Sahara" is a big surprise. Information theory measures surprise with a number, and entropy is the average surprise of a random event.

Cross entropy and KL divergence Suppose reality follows p, but your model believes q. You will be surprised more often than necessary. Cross entropy measures your average surprise, and the extra over the unavoidable entropy is the KL divergence .

Softmax, temperature and log-sum-exp A classifier first produces raw scores called logits , which can be any real numbers. Softmax turns them into probabilities: all positive and adding to 1, with the biggest logit getting the biggest share.

Loss functions for regression A loss function turns "how wrong was the model?" into one number that training then makes smaller. The choice of loss decides what kind of mistakes the model cares most about.

Log loss for classification For a yes or no classifier that outputs a probability p, the standard loss is log loss , also called binary cross entropy. It is the cross entropy from earlier with only two classes.

Regularisation: L2 and L1 A flexible model can fit noise (overfitting, Chapter 11). Regularisation adds a penalty for large weights to the loss, so the model only uses big weights when the data really demands them.

Classification metrics, ROC and AUC Accuracy alone can fool you. If 99% of card payments are honest, a model that never flags fraud is 99% accurate and completely useless. The confusion matrix shows what really happened.

Surprise (information) of an outcome
$I(x) = \log_2 \dfrac{1}{p(x)} = -\log_2 p(x)$

Cross entropy $H(p, q) = -\Sigma\, p(x) \log q(x)$

Softmax $\text{softmax}(z)_i = \dfrac{e^{z_i \div T}}{\Sigma_j\, e^{z_j \div T}}$

Mean squared error
$MSE = \dfrac{1}{n}\,\Sigma\,(y_i - \hat{y}_i)^2$

Binary cross entropy
$L = -[\,y \log p + (1-y)\log(1-p)\,]$

L2 (ridge, weight decay) $\text{loss} + \lambda\,\Sigma\,w_j^2$

Precision $\dfrac{TP}{TP + FP}$

TRAP Why not use MSE for classification? With a sigmoid output, MSE gives tiny gradients exactly when the model is confidently wrong, so learning stalls. Log loss punishes confident mistakes hard and keeps the gradient $p - y$ large.

STEP 6 Advanced maths for ML

6.1 Orthogonality, Matrix Decompositions and Tensors

Orthogonal vectors and orthonormal bases Two vectors are orthogonal when they meet at a right angle, which happens exactly when their dot product is 0 (Chapter 3). A set of vectors that are all orthogonal to each other and all have length 1 is called orthonormal .

Gram Schmidt and the QR decomposition Most real bases are not orthonormal. The Gram Schmidt process turns any set of independent vectors into an orthonormal set that spans exactly the same space.

Quadratic forms and positive definite matrices A quadratic form is the matrix version of ax^2. For a symmetric matrix A and a vector x, the number x^TAx is a sum of squares and cross terms. In two dimensions it is a surface over the plane, and its shape is decided by A.

LU and Cholesky: solving systems fast Gaussian elimination (Chapter 4) secretly builds a factorisation. Record the multipliers you used to clear each column, put them in a lower triangular matrix L with 1s on the diagonal, and keep the staircase you ended with as an upper triangular matrix U.

Least squares as a projection When there are more equations than unknowns, as with 1000 houses and 3 weights, $X\beta = y$ has no exact answer. Least squares picks the β whose predictions $X\beta$ are as close as possible to y. There is a beautiful picture behind it.

Matrix norms, trace and the condition number Vectors have lengths (Chapter 3). Matrices need a size too, for regularising weight matrices, for measuring how far apart two matrices are, and for judging how sensitive a problem is. There are a few standard choices.

Tensors, shapes and broadcasting A tensor in machine learning code is simply an array with any number of directions, called axes or dimensions. A number is a 0D tensor, a vector is 1D, a matrix is 2D, and so on. The shape lists the size along each axis.

Orthonormal
$q_i \cdot q_j = 1$ if i = j, and 0 if $i \neq j$

QR decomposition $A = Q\,R$

Quadratic form in 2D
x^TAx with A = [[a, b], [b, c]] gives $a\,x_1^2 + 2b\,x_1x_2 + c\,x_2^2$

LU decomposition $A = L\,U$

The residual is perpendicular to every column

$$X^T(y - X\beta) = 0 \implies X^TX\beta = X^Ty$$

Rule of thumb for lost digits

$$\text{digits lost} \approx \log_{10} \kappa$$

6.2 Taylor Series, Newton's Method and Automatic Differentiation

Taylor series: curves as polynomials Computers can only add and multiply. To find e 0.5 or sin 0.3, a calculator uses a polynomial that copies the function very closely near one point. That copy is a Taylor polynomial .

Newton's method for roots and minimums To solve f(x) = 0, Newton's method draws the tangent line at your current guess and jumps to the point where that line crosses zero. The tangent line is the order 1 Taylor polynomial, so each step solves a simpler, straight line version of the problem.

Matrix calculus: gradients without index pain When a function takes a whole vector or matrix as input, we want its gradient as a vector or matrix of the same shape.

Automatic differentiation: forward and reverse mode There are three ways a computer can find derivatives. Numerical differences, (f(x + h) − f(x)) ÷ h, are easy but slow and slightly inaccurate. Symbolic differentiation, like a computer algebra system, produces exact formulas that can grow huge.

Vanishing and exploding gradients Backpropagation through a deep network multiplies many factors together, one per layer. Each factor is roughly (slope of the activation) × (weight). If those factors are a little below 1, their product shrinks towards 0 and the early layers stop learning.

Convexity and Jensen's inequality A function is convex when the straight line (chord) between any two points on its graph never goes below the graph. Think of the shape of a bowl.

Numerical stability: when maths meets floating point Floating point numbers (Chapter 1) have about 16 significant digits in float64 and about 7 in float32. Formulas that are perfect on paper can lose all their accuracy on a computer. Three situations come up again and again in machine learning.

Taylor polynomial around 0 (also called a Maclaurin polynomial)

$$f(x) \approx f(0) + f'(0)\,x + \frac{f''(0)}{2!}\,x^2 + \frac{f'''(0)}{3!}\,x^3 + \dots$$

Newton step for a root

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Gradient reaching layer 1 of an L layer chain

$$\partial L/\partial h_1 = \partial L/\partial h_L \times \prod (\sigma'(z_k)\, w_k)$$

Definition

$$f(t\,a + (1 - t)\,b) \le t\,f(a) + (1 - t)\,f(b) \text{ for all } t \text{ in } [0, 1]$$

The log-sum-exp trick

$$\ln \Sigma\, e^{z_i} = m + \ln \Sigma\, e^{z_i - m} \text{ with } m = \max z_i$$

6.3 Bayesian Inference, Markov Chains and Sampling

Joint, marginal and conditional distributions Real data has several variables at once: the weather and whether people carry umbrellas, a patient's age and blood pressure, a pixel and its neighbour. The joint distribution P(X, Y) gives the chance of every combination. From it you can get everything else.

The multivariate normal and Mahalanobis distance The bell curve of Chapter 10 has a version for many variables at once, the multivariate normal . It is described by a mean vector μ (the centre) and a covariance matrix Σ (the spread of each variable and how pairs move together).

Bayesian inference: priors, posteriors and MAP Chapter 9 used Bayes' rule for events. Bayesian inference uses it for an unknown parameter , such as the chance θ that a coin lands heads. Instead of one best guess, you keep a whole distribution of belief over θ and update it with data.

Markov chains and stationary distributions A Markov chain moves between states at random, and the chance of the next state depends only on the current one, not on the history. Weather from day to day, a shopper moving between web pages and the next word in a very simple text model are common examples.

Monte Carlo and inverse transform sampling Monte Carlo methods answer questions by simulation. To find an average, a probability or an integral, draw many random samples and average the results.

MCMC and the Metropolis algorithm Bayesian posteriors are often known only up to a constant: we can compute likelihood × prior, but not p(data), which needs an impossible integral. Markov chain Monte Carlo (MCMC) still produces samples.

Concentration inequalities and the law of large numbers The law of large numbers says that the average of many independent samples gets close to the true mean. Concentration inequalities say how close , with a guarantee rather than an approximation.

Marginal (sum out the other variable)

$$P(X = x) = \Sigma_y\, P(X = x, Y = y)$$

Density of a d dimensional normal

$$p(x) = \frac{1}{(2\pi)^{d/2}\,|\Sigma|^{1/2}} \exp(-\tfrac{1}{2}\,(x - \mu)^T\,\Sigma^{-1}\,(x - \mu))$$

Bayes' rule for a parameter

$$p(\theta \mid \text{data}) = \frac{p(\text{data} \mid \theta) \cdot p(\theta)}{p(\text{data})} \propto \text{likelihood} \times \text{prior}$$

Transition matrix

$P_{ij} = P(\text{next} = j \mid \text{now} = i)$, each row adds up to 1

Monte Carlo estimate

$$E[f(X)] \approx \frac{1}{n}\,\Sigma\, f(x_i), \text{ with } x_i \text{ drawn from } X$$